

SULIT



SECP2523 DATABASE

SESSION 2024/2025 – SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

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Case Study	Improvement of KKK Online System
System Name	KKK Online System
Group Name	Tech Hi Five

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SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

The KKK Online System is a system proposed to substitute the manual system used by Koperasi Kakitangan KADA (KKK) Berhad. The cooperative is solely for the employees in Lembaga Kemajuan Pertanian Kemubu (KADA).

The cooperative is currently using a manual system to handle daily operations such as member application, loan application and salary deduction. This manual process will cause inconvenience to the members and staffs as they need to fill in physical forms and do manual calculation. Besides, the members are not able to view their financial information online. Thus, a digital system is proposed.

The digital system has digitalised the manual process and adds some new functionality to the system. The KKK Online System can allow KADA employees to apply membership via the system. Upon approval, the applicant will receive an email and officially become a member of the cooperative. A member can view their financial status associated with the cooperative, edit their profile, and apply for financing services.

An admin on the other hand, can view the list of member applications and loan applications, and update their status respectively. The results of the applications are decided during the meeting of the Board of Directors. Besides, the admin can view the list of existing members and list of loans that are not settled. On top of that, an admin can modify the financial status. Once the cooperative has received the money for monthly deduction from respective employers, the admin can select the members involved and updated their financial statuses by batch processing. The admin can also record the transaction one by one for ad-hoc or additional transactions. The admin can also update the policies such as the fee amount for member application, maximum financing amount, loan interest rate and monthly salary deduction amount. The information of the policies will be reflected in member application, loan application and salary deduction. Furthermore, the admin can update the website banner that is displayed at the main page of the member. Last but not least, an admin can generate monthly and yearly summary report.

Admin Updating Module is assigned to me. This module mainly manages the system and updates various important data. The functionalities of this module are updating policies, updating banners and record transactions.

Group Members' Name	Matric No	Module
Tan Yi Ya	A23CS0187	User module - Authentication - Member Application
Chua Jia Lin	A23CS0069	Member module - View financial status - Update user profile (Not in a module) - Admin dashboard
Teh Ru Qian	A23CA0191	Loan Module - Loan application Admin Reporting Module - Generate report
Goe Jie Ying	A23CS0224	Admin Approval Module - Approve membership - Approve loan application - View member list - View loan list
Lam Yoke Yu (Me)	A23CS0233	Admin Updating Module - Manage System

Table 1 Module distribution

1.2 Project Weaknesses or Improvement

First and foremost, there is no filter and search in the monthly salary deduction feature. The admin would need to go page by page to find the specific member. Figure 1 below shows the snippet of the current salary deduction transaction page. Thus, a search feature and a filter feature can be added to the page.

☐ Select All	No Ahli	Nama	Modah Syer	Modal Yuran	Simpanan Tetap	Tabung Anggota	Simpanan Anggota	Al-Bai	Al-Innah	Baik Pulih Kenderaan	Road Tax dan Insurans	Khas	Karnival Musim Istimewa	Al-Qadrul Hassan	Dikemaskini
☐	1000	Roslan bin Roslan	400.00	0.00	50.00	20.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1001	Pravin A/L Ali	200.00	0.00	50.00	20.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025
☐	1003	Lam Kin Xin	300.00	0.00	150.00	20.00	50.00	0.00	0.00	13,361.66	0.00	0.00	0.00	0.00	21-01-2025
☐	1004	Farra bin	150.00	0.00	0.00	15.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21-01-2025

Figure 1 Salary Deduction Transaction Page of the Proposed System

For additional transactions (Transaksi Lain), there may be also lack of robustness in the information collected. The admin can simply make the change without needing to upload any transaction proof. Figure 2 shows snippet of the transaction page. A field may be added to the page to upload the transaction proof.

Transaksi Lain

No Anggota

Maklumat Anggota

Name	Roslan bin Roslan
IC	010101-01-0101
No. Anggota	1000
No. PF	1005

Maklumat Saham Ahli

	Status Semasa	Perubahan	Status Baru
Modal Syer	400.00	0.00	400.00
Modal Yuran	0.00	0.00	0.00
Simpanan Tetap	50.00	0.00	50.00
Simpanan Anggota	0.00	0.00	0.00
Tabung Anggota	20.00	0.00	20.00

Jumlah Bayaran: RM 0.00

Ulasan

Hantar

Figure 2 Additional Transaction Page of the Proposed System

Other than that, the policies that are included in the update policies page may not be robust enough due to the lack of information from the client. Given that we included the all the fields for the financing requirements, as shown in Figure 3 and 4 and the monthly salary deduction requirements, as shown in Figure 5 and 6, we may add more field to the functionality to increase flexibility of the system where the policy may change in the future. This may include the financing rate of each type of loan specifically and their maximum financing amount for each type of loan and the monthly salary deduction amount for each type of financial statuses.

Polisi Permohonan Pembiayaan

Kemaskini

Modal Syer Minimum Peminjam	RM 300.00					
Kadar Keuntungan	4.20 %					
Tempoh Ansuran Maksima	6 tahun					
Jumlah Pembiayaan Maksima	RM 20,000.00					
Jadual Pembayaran Balik Pembiayaan						
Kadar Keuntungan	4.20 %					
Tempoh (Tahun)	1	2	3	4	5	6
Tempoh (Bulan)	12	24	36	48	60	72
Jumlah Pembiayaan	Ansuran Bulanan					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78

Figure 3 Policies of the Financing Application

Permohonan Anggota
Permohonan Pinjaman
Senarai Anggota
Senarai Peminjam
Transaksi
Kemaskini Polisi
Kemaskini Iklan
Laporan

Kemaskini Polisi Permohonan Pembiayaan

Modal Syer Minimum Peminjam

RM
300
.00

Kadar Keuntungan

4.2
%

Tempoh Ansuran Maksima

6
tahun

Jumlah Pembiayaan Maksima

RM
20000
.00

Figure 4 Form to Update the Financing Policies

Polisi Potongan Gaji		Kemaskini
Simpanan Tetap		RM 50.00
Sumbangan Tabung Kebajikan		RM 5.00

Figure 5 Policies for Monthly Salary Deduction

Permohonan Anggota Permohonan Pinjaman Senarai Anggota Senarai Peminjam Transaksi ▾ Kemaskini Polisi Kemaskini Iklan Laporan

Kemaskini Polisi Potongan Gaji

Simpanan Tetap

RM 50 .00

Sumbangan Tabung Kebajikan

RM 5 .00

Kembali Kemaskini

Figure 6 Form to Update the Monthly Salary Deduction Amount

All changes made by the admin for policies and transaction is logged in the database for checking, however, there is no page that display the changes and lead to integrity issues. Thus, a function can be added to show the change history.

Finally, there are no information recorded for admin other than the admin ID and password. Besides, we cannot register new admin to the system. Thus, a new functionality can be added to the system.

1.3 Proposed Module

Based on the weakness and improvements identified in Section 1.2, the functionalities and features mentioned, the current module should be renamed from admin updating module to administrator module, as it handles more task than updating.

On top of the existing functionality, the module should

- Add filters and search feature in the salary deduction page
- Add fields to upload proof for the additional transactions (this one done)
- Add fields to the update policies page to increase robustness and flexibility

For the new functionalities and pages, the module should allow the admin to

- View the policy changes
- View transaction log (done)
- Add more information of admin to the system
- Register new admins to the system

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD

Based on the existing conceptual ERD, new attributes are added to the entities.

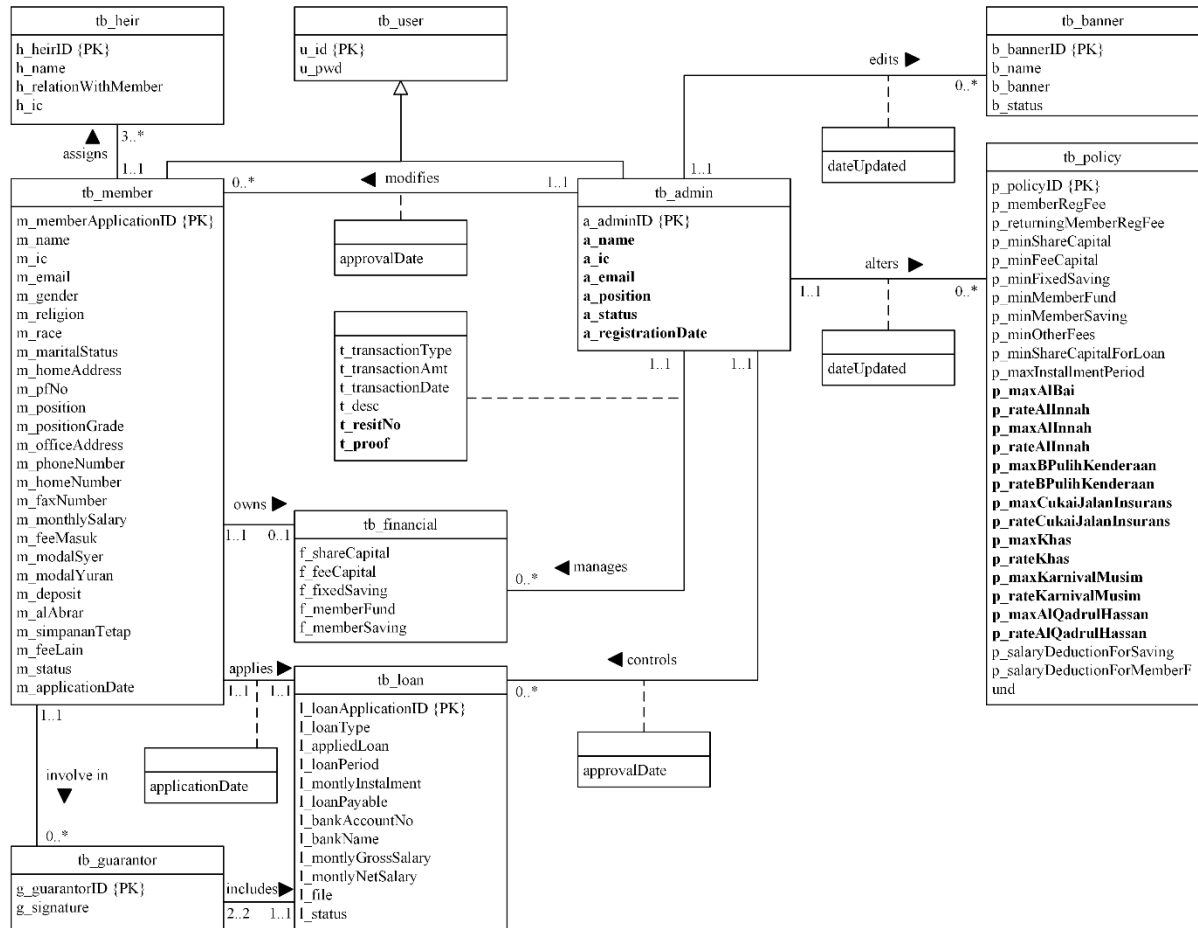
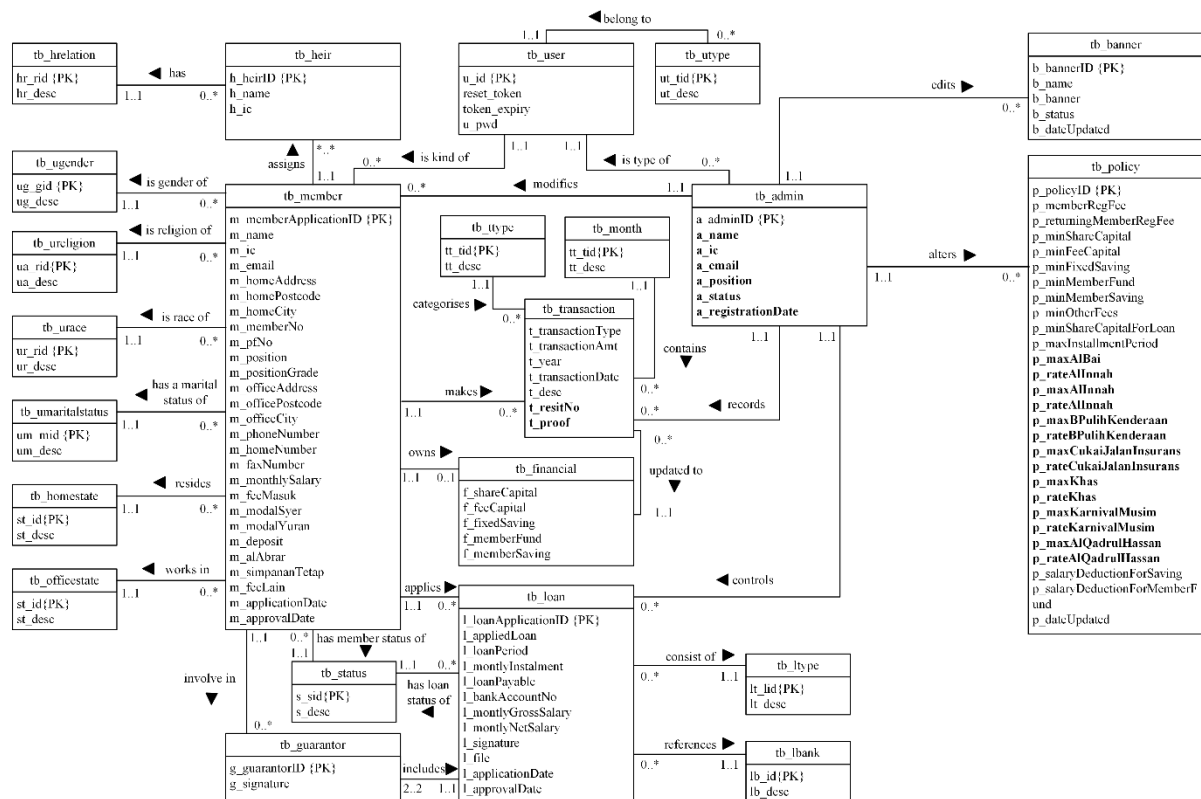


Figure 7 Conceptual ERD for the Proposed Module

2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

2.3.1 Description of entities

Entity name	Description	Aliases	Occurrence
tb_user	General term describing all users using KKK Online System	User	Admin and member are types of user. Each user has one user type.
tb_utype	General term describing all types of users using KKK Online System	User type	Each user type belongs to many users.
tb_admin	General term describing all user logged in as admin using KKK Online System	KKK Staff	Admin is one type of user. Admin can edit many banners. Each admin alters many policies. Each admin controls many loan applications. Each admin approves many member applications. Each admin can record many transaction records.

Entity name	Description	Aliases	Occurrence
tb_member	General term describing all user logged in as member using KKK Online System	KKK member, Member application	Each member has one gender, one religion, one race and one marital status only. Each member resides and works in one state. Each member can become guarantors for other members. Each member can apply for many loans. Each member application will have one status only at any one time. Each member owns at most one financial record. Each member application is approved by one admin. Each member can make many transactions. Members are one kind of user for the system. Each member can assign many heirs.
tb_ugender	General term describing the gender of a member	Gender	Each gender can be shared by many members.
tb_ureligion	General term describing the religion of a member	Religion	Each religion can be shared by many members.
tb_urace	General term describing the race of a member	Race	Each race can be shared by many members.
tb_umaritalstatus	General term describing the marital status of a member	Marital status	Each marital status can be shared by many members.
tb_homestate	General term describing the home state of a member	State	There can be none or many members residing in the state.
tb_officestate	General term describing the office state of a member	State	There can be none or many members working in the state.
tb_heir	General term describing all people declared as heir of a member applicant	Heir	Each heir is assigned to one member. Each heir for a single member will have a relationship with members only.

Entity name	Description	Aliases	Occurrence
tb_hrelation	General term describing the relationship between a member applicant and his heir	Relationship	Each heir relation with a member may be having by many members.
tb_banner	General term describing all banners or picture to be published on Member Main Page	-	Each banner is edited by one admin.
tb_policies	General term describing all policies of the cooperative	-	Each policy is altered by a single admin only.
tb_transaction	General term describing all transaction made by member	-	Each transaction has one transaction type only. Each transaction record occurs in one month. Each transaction record will be recorded by a single admin only. Each transaction made by one member. Each transaction record will be updated to a member's financial record only.
tb_ttype	General term describing all types of transactions	Transaction type	Each transaction type belongs to many transactions.
tb_month	General term describing all month in a year	Month	Each month may be contained in one or more transaction records.
tb_financial	General term describing all financial status of a member	Financial status	Each financial record is owned by one member only. Each financial record can be updated by many transaction records.
tb_loan	General term describing all loans of the cooperative	Loan Application	Each loan is controlled by one admin. Each loan consists of one loan type. Each loan references a single bank only. Each loan application has a status at one time. Each loan application is applied by one member. Each loan application required two guarantors.

Entity name	Description	Aliases	Occurrence
tb_ltype	General term describing all types of loans of the cooperative	Loan Type	Each loan type is being referenced by one or more loan applications.
tb_lbank	General term describing all banks	Bank	Each bank is being referenced by one or more loan applications.
tb_guarantor	General term describing all guarantors for a loan applicant	-	Two guarantors are required for each loan application. Each guarantor is involved by one member only.
tb_status	General term describing all status for the applications	-	Each membership and loan application has a single status at any one time.

Table 2 Data Dictionary for Administrator Module showing a description of entities

2.3.2 Description of relationships

Entity name	Multiplicity	Relationship	Entity name	Multiplicity
tb_user	1..1	is having	tb_ctype	1..*
tb_admin	1..1	edits	tb_banner	0..*
	1..1	alters	tb_policies	0..*
	1..1	controls	tb_loan	0..*
	1..1	record	tb_financial	0..*
	1..1	modifies	tb_member	0..*
tb_member	0..*	is a type of	tb_user	1..1
	1..*	assigns	tb_heir	*..*
	1..1	involve in	tb_guarantor	0..*
	1..1	applies	tb_loan	0..*
	1..1	owns	tb_financial	0..1
	1..1	makes	tb_transaction	0..*
	0..*	is gender of	tb_ugender	1..1
	0..*	is religion of	tb_uireligion	1..1
	0..*	is race of	tb_urace	1..1
	0..*	has a marital status of	tb_umaritalstatus	1..1
	0..*	resides	tb_homestate	1..1
	0..*	works in	tb_officestate	1..1
	0..*	has a member application status of	tb_status	1..1
	0..*	is a kind of	tb_user	1..1
tb_heir	0..*	has	tb_hrelation	1..1
tb_loan	1..1	includes	tb_guarantor	2..2
	0..*	consists of	tb_ltype	1..1
	0..*	references	tb_lbank	1..1
	0..*	has a loan status of	tb_status	1..1
tb_transaction	0..*	categorizes	tb_ttype	1..1
	0..*	updated to	tb_financial	1..1
	0..*	contains	tb_month	1..1

Table 3 Data Dictionary for Administrator Module showing a description of relationships

2.3.3 Description of attributes

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
tb_user	u_id (PK)	int	Uniquely identifies each user	No	1
	reset_token	varchar (255)	Used when user is going to reset the password	Yes	1440f2ccd0064f390b491ca7caec30b5
	token_expiry	timestamp	Record the time when user reset password	Yes	2025-11-11 16:00:00
	u_pwd	varchar (255)	User password for login	No	Abc@123
tb_admin	a_adminID (PK)	int	Uniquely identifies each admin	No	12345678
	a_name	varchar (50)	Admin's full name	No	Ali bin Abu
	a_ic	varchar (14)	Admin's identification number	No	850101-08-1033
	a_email	varchar (255)	Admin's email address	No	ali@gmail.com
	a_position	varchar (50)	Admin's position	No	Pengurus
	a_status	tinyint (1)	Admin's active or inactive state	No	0
	a_registrationDate	timestamp	Date where admin is registered	No	2025-01-15 06:40:00
tb_utype	ut_tid (PK)	int	Uniquely identifies each user type	No	1
	ut_desc	varchar (10)	User's type description	No	Admin
tb_member	memberApplicationID (PK)	int	Uniquely identifies each member application	No	1001
	m_name	varchar (50)	User's full name	No	Ali bin Abu
	m_ic	varchar (14)	User's identification number	No	850101-08-1033
	m_email	varchar (255)	User's email address	No	ali@gmail.com
	m_homeAddress	varchar (255)	User's home address	No	1, Taman Bahagia, Jalan Bahagia,
	m_homePostcode	int	User's home postcode	No	57000
	m_homeCity	varchar (50)	User's home city	No	Kuala Lumpur

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	m_memberNo	int	Uniquely identifies each member	No	1
	m_pfNo	int	User's PF number	No	1001
	m_position	varchar (50)	User's position	No	Pengurus
	m_positionGrade	varchar (11)	User's position grade	No	G41
	m_officeAddress	varchar (255)	User's office address	No	25, Jalan Makmur
	m_officePostcode	int	User's office postcode	No	57000
	m_officeCity	varchar (50)	User's office city	No	Kuala Lumpur
	m_phoneNumber	varchar (13)	User's phone number	No	0123456789
	m_homeNumber	varchar (13)	User's home number	Yes	051234567
	m_monthlySalary	double	User's monthly salary	No	3500.00
	m_feeMasuk	double	User's entry fee	No	45.00
	m_modalShare	double	User's shares	No	300.00
	m_modalYuran	double	User's share fee	No	50.00
	m_deposit	double	User's deposit	No	50.00
	m_alAbrar	double	A kind of user's fund (Sumbangan Tabung Kebajikan Anggota)	No	5.00
	m_simpanan Tetap	double	User's fixed saving	No	0.00
	m_feeLain	double	User's other fee	No	0.00
	m_applicationDate	datetime	The date when user submit his/her member application	No	2025-01-03 16:00:00
	m_approvalDate	timestamp	The date when the member application being approved by admin	Yes	2025-01-15 06:40:00
tb_ugender	ug_gid (PK)	int	The unique id for gender	No	1
	ug_desc	varchar (10)	User's gender description	No	Lelaki
tb_ureligion	ua_rid (PK)	int	The unique id for religion	No	1
	ua_desc	varchar (10)	User's religion description	No	Islam
tb_urace	ur_rid (PK)	int	The unique id for race	No	1
	ur_desc	varchar (10)	User gender's description	No	Lelaki
tb_umaritalstatus	um_mid (PK)	int	The unique id for marital status	No	1

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	um_desc	varchar (10)	User's marital status description	No	Bujang
tb_home state	st_id (PK)	int	The unique id for user's home state	No	1
	st_desc	varchar (15)	User's home state description	No	Johor
tb_office state	st_id (PK)	int	The unique id for user's office state	No	1
	st_desc	varchar (15)	User's office state description	No	Melaka
tb_heir	h_heirID (PK)	int	Uniquely identifies each heir	No	1
	h_name	varchar (50)	Heir's full name	No	Tan Mei Ling
	h_ic	varchar (14)	Heir's identification number	No	010101-01-0101
tb_hrelation	hr_rid (PK)	int	Uniquely identifies each heir relationship with member	No	1
	hr_desc	varchar (20)	The description of heir relationship with member	No	Suami Isteri
tb_banner	b_bannerID (PK)	int	Uniquely identifies each banner	No	1
	b_name	varchar (255)	Banner's name	No	Sunny 16 boat
	b_banner	varchar (50)	Banner's picture	No	advertisement.jpg
	b_status	tinyint (1)	Banner's active or inactive state	No	0
	b_dateUpdated	timestamp	Banner's updated time	No	2025-01-19 11:31:43
tb_policies	p_policyID (PK)	int	Uniquely identifies each policy	No	1
	p_memberRegFee	double	Policy for member registration fee	No	35.00
	p_returningMemberRegFee	double	Policy for the amount of member registration fee for returning member	No	100.00
	p_minShareCapital	double	Policy for minimum share capital of member applicant	No	300
	p_minFeeCapital	double	Policy for minimum fee capital of member applicant	No	0
	p_minFixedSaving	double	Policy for minimum fixed saving of member applicant	No	0
	p_minMemberFund	double	Policy for minimum member fund of member applicant	No	5

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	p_minMemberSaving	double	Policy for minimum member saving of member applicant	No	50
	p_minOther Fees	double	Policy for minimum other fees of member applicant	No	0
	p_minShare CapitalForLoan	double	Policy for minimum share capital loan of loan applicant	No	300
	p_maxInstallmentPeriod	int	Policy for maximum instalment period of loan applicant	No	6
	p_maxAlBai	double	Policy for maximum financing amount for Al-Bai loan	No	20000
	p_rateAlInnah	double	Policy for interest rate for Al-Bai loan	No	4.2
	p_maxAlInnah	double	Policy for maximum financing amount for Al-Innah loan	No	20000
	p_rateAlInnah	double	Policy for interest rate for Al-Innah loan	No	4.2
	p_maxBPulihKendaraan	double	Policy for maximum financing amount for Baik Pulih Kendaraan loan	No	4500
	p_rateBPulihKendaraan	double	Policy for interest rate for Baik Pulih Kendaraan loan	No	4.2
	p_maxCukaiJalanInsurans	double	Policy for maximum financing amount for Road Tax and Insurance loan	No	4500
	p_rateCukaiJalanInsurans	double	Policy for interest rate for Road Tax and Insurance loan	No	4.2
	p_maxKhas	double	Policy for maximum financing amount for Special Scheme loan	No	10000
	p_rateKhas	double	Policy for interest rate for Special Scheme loan	No	4.2
	p_maxKarnivalMusim	double	Policy for maximum financing amount for Special Season Carnival loan	No	10000
	p_rateKarnivalMusim	double	Policy for interest rate for Special Season Carnival loan	No	4.2
	p_maxAlQadrulHassan	double	Policy for maximum financing amount for Al-Qadrul Hassan loan	No	20000

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	p_rateAlQadrulHassan	double	Policy for interest rate for Al-Qadrul Hassan loan	No	4.2
	p_salaryDeductionForSaving	double	Policy for salary deduction for saving of loan borrower	No	50
	p_salaryDeductionForMemberFund	double	Policy for salary deduction for member fund of loan borrower	No	5
	p_dateUpdated	timestamp	The date when the admin update the policy	No	2024-12-20 17:30:00
tb_transaction	t_transaction ID (PK)	int	Uniquely identifies each transaction	No	1
	t_transaction Amt	double	Transaction amount	No	100
	t_year	int	The year when the transaction made	No	2024
	t_transaction Date	timestamp	Transaction date	No	2024-12-20 17:30:00
	t_desc	varchar (255)	The objective of the transaction made	No	Potongan Gaji Bayaran Balik 8
	t_resitNo	varchar (255)	Receipt number of the transaction	No	12456789
	t_proof	varchar (255)	Picture or document of the receipt	No	transaksi_tambahan_1737620932.jpg
tb_ttype	tt_lid (PK)	int	Uniquely identifies each transaction type	No	1
	tt_desc	varchar (30)	The description of each transaction type	No	Al-Qadrul Hassan
tb_financial	f_shareCapital	double	Member's share capital	No	10
	f_feeCapital	double	Member's fee capital	No	100
	f_fixedSaving	double	Member's fixed saving	No	200
	f_memberFund	double	Member's fund	No	50
	f_memberSaving	double	Member's saving	No	200

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	f_dateUpdated	double	The date when the financial record is updated	No	2025-01-21 15:36:25
tb_loan	l_loanApplicationID (PK)	int	Uniquely identifies each loan application	No	No
	l_appliedLoan	double	Amount of loan applied	No	No
	l_loanPeriod	int	Loan period	No	No
	l_monthlyInstalment	double	Monthly instalment for loan	No	No
	l_loanPayable	double	The total loan that need to be paid by the member	No	No
	l_bankAccountNo	int	Loan borrower's bank account number	No	214748364
	l_monthlyGrossSalary	double	Loan borrower's monthly gross salary	No	10000
	l_monthlyNetSalary	double	Loan borrower's monthly net salary	No	8000
	l_signature	varchar (50)	Loan borrower's signature	No	./uploads/8d969f4e9e3586a1226cc389a6a29b1b.jpg
	l_file	varchar (50)	Loan borrower's employer's verification proof	No	pengesahan_majikan_4_1737462659.pdf
	l_applicationDate	datetime	The date where loan application is made by a member	No	2025-01-05 12:51:36
	l_approvalDate	timestamp	The date where loan application is approved by admin	No	2025-01-11 23:40:25
tb_ltype	lt_lid (PK)	int	Uniquely identifies each retrieval record	No	1
	lt_desc	varchar (30)	The description of each loan type	No	Al-Qadrul Hassan
tb_lbank	lb_lid (PK)	int	Uniquely identifies each bank	No	1
	lb_desc	varchar (30)	The description of each bank	No	BSN
tb_guarantor	g_guarantorID (PK)	int	Uniquely identifies each guarantor	No	1

Entity name	Attributes	Data Type & Length	Description	Nulls	Example
	g_signature	varchar (50)	Guarantor's signature	No	./uploads/678a24a16eee9.jpg
tb_status	s_sid (PK)	int	Uniquely identifies each status	No	2
	s_desc	varchar (20)	The description of each status	No	Ditolak
tb_rmont h	rm_id (PK)	int	Uniquely identifies each month	No	3
	rm_desc	varchar (10)	The description of each month	No	Mac

Table 4 Data Dictionary for Administrator Module showing a description of attributes

2.4 Relational Database Schema

1st Normal Form

tb_user (u_id, reset_token, token_expiry, u_pwd, ut_tid, ut_desc)
PK: u_id

tb_admin (a_adminID, a_name, a_ic, a_email, a_position, a_status, a_registrationDate)
FK: a_adminID references tb_user (u_id)

tb_member (m_memberApplicationID, m_name, m_ic, m_email, ug_gid, ug_desc, ua_rid, ua_desc, ur_rid, ur_desc, um_mid, um_desc, m_homeAddress, m_homePostcode, m_homeCity, st_id, st_desc, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, st_id, st_desc, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, s_sid, s_desc, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (a_adminID)

tb_heir (h_heirID, h_memberApplicationID, h_name, hr_rid, hr_desc, h_ic)
PK: h_heirID
FK: h_memberApplicationID references tb_member (m_memberApplicationID)

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
FK: f_memberNo references tb_member (m_memberNo)

tb_banner (b_bannerID, b_name, b_banner, b_status, b_dateUpdated, b_adminID)
PK: b_bannerID
FK: b_adminID references tb_admin (a_adminID)

tb_loan (l_loanApplicationID, l_memberNo, lt_lid, lt_desc, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, lb_id, lb_desc, l_monthlyNetSalary, l_signature, l_file, s_sid, s_desc, l_applicationDate, l_approvalDate, l_adminID)

PK: l_loanApplicationID
FK: l_memberNo references tb_member (m_memberNo)
FK: l_adminID references tb_admin (a_adminID)

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)
PK: g_guarantorID
FK: g_loanApplicationID references tb_loan (l_loanApplicationID)
FK: g_memberNo references tb_member (m_memberNo)

tb_policies (p_policyID, p_memberRegFee, p_minShareCapital, p_minFeeCapital, p_minFixedSaving, p_minMemberFund, p_minMemberSaving, p_minOtherFees, p_minShareCapitalForLoan, p_maxInstallmentPeriod, p_maxAlBai, p_rateAlInnah, p_maxAlInnah, p_rateAlInnah, p_maxBPulihKenderaan, p_rateBPulihKenderaan, p_maxCukaiJalanInsurans, p_rateCukaiJalanInsurans, p_maxKhas, p_rateKhas, p_maxKarnivalMusim, p_rateKarnivalMusim, p_maxAlQadrulHassan, p_rateAlQadrulHassan, p_salaryDeductionForSaving, p_salaryDeductionForMemberFund, p_dateUpdated, p_adminID)
PK: p_policyID
FK: p_adminID references tb_admin (a_adminID)

tb_transaction (t_transactionID, tt_id, tt_desc, t_transactionAmt, rm_id, rm_desc, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)
PK: t_transactionID
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

2.5 Normalisation

2nd Normal Form

No partial dependency identified as there is no 1NF relation with composite PK.

tb_user (u_id, reset_token, token_expiry, u_pwd, ut_tid, ut_desc)

PK: u_id

tb_admin (a_adminID, a_name, a_ic, a_email, a_position, a_status, a_registrationDate)

FK: a_adminID references tb_user (u_id)

tb_member (m_memberApplicationID, m_name, m_ic, m_email, ug_gid, ug_desc, ua_rid, ua_desc, ur_rid, ur_desc, um_mid, um_desc, m_homeAddress, m_homePostcode, m_homeCity, st_id, st_desc, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, st_id, st_desc, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, s_sid, s_desc, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID

FK: m_adminID references tb_admin (a_adminID)

tb_heir (h_heirID, h_memberApplicationID, h_name, hr_rid, hr_desc, h_ic)

PK: h_heirID

FK: h_memberApplicationID references tb_member (m_memberApplicationID)

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)

FK: f_memberNo references tb_member (m_memberNo)

tb_banner (b_bannerID, b_name, b_banner, b_status, b_dateUpdated, b_adminID)

PK: b_bannerID

FK: b_adminID references tb_admin (a_adminID)

tb_loan (l_loanApplicationID, l_memberNo, lt_lid, lt_desc, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, lb_id, lb_desc, l_monthlyNetSalary, l_signature, l_file, s_sid, s_desc, l_applicationDate, l_approvalDate, l_adminID)

PK: l_loanApplicationID

FK: l_memberNo references tb_member (m_memberNo)

FK: l_adminID references tb_admin (a_adminID)

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)

PK: g_guarantorID

FK: g_loanApplicationID references tb_loan (l_loanApplicationID)

FK: g_memberNo references tb_member (m_memberNo)

tb_policies (p_policyID, p_memberRegFee, p_minShareCapital, p_minFeeCapital, p_minFixedSaving, p_minMemberFund, p_minMemberSaving, p_minOtherFees, p_minShareCapitalForLoan, p_maxInstallmentPeriod, p_maxAlBai, p_rateAlInnah, p_maxAlInnah, p_rateAlInnah, p_maxBPulihKenderaan, p_rateBPulihKenderaan, p_maxCukaiJalanInsurans, p_rateCukaiJalanInsurans, p_maxKhas, p_rateKhas,

p_maxKarnivalMusim, p_rateKarnivalMusim, p_maxAlQadrulHassan,
p_rateAlQadrulHassan, p_salaryDeductionForSaving, p_salaryDeductionForMemberFund,
p_dateUpdated, p_adminID)

PK: p_policyID

FK: p_adminID references tb_admin (a_adminID)

tb_transaction (t_transactionID, tt_id, tt_desc, t_transactionAmt, rm_id, rm_desc, t_year,
t_desc, t_transactionDate, t_memberNo, t_adminID)

PK: t_transactionID

FK: t_memberNo references tb_member (m_memberNo)

FK: t_adminID references tb_admin (a_adminID)

3rd Normal Form

tb_user (u_id, reset_token, token_expiry, u_pwd, ut_tid, ut_desc)
PK: u_id

Transitive Dependency:
FD1: ut_tid → ut_desc

tb_user (u_id, reset_token, token_expiry, u_pwd, u_type)
PK: u_id
FK: u_type references tb_utype (ut_tid)

tb_utype (ut_tid, ut_desc)
PK: ut_tid

tb_admin (a_adminID, a_name, a_ic, a_email, a_position, a_status, a_registrationDate)
FK: a_adminID references tb_user (u_id)

No Transitive dependency.

tb_member (m_memberApplicationID, m_name, m_ic, m_email, ug_gid, ug_desc, ua_rid, ua_desc, ur_rid, ur_desc, um_mid, um_desc, m_homeAddress, m_homePostcode, m_homeCity, st_id, st_desc, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, st_id, st_desc, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, s_sid, s_desc, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_adminID references tb_admin (a_adminID)

Transitive Dependency:
FD1: ug_gid → ug_desc (Partial Dependency)
FD2: ua_rid → ua_desc (Partial Dependency)
FD3: um_mid → um_desc (Partial Dependency)
FD4: st_id → st_desc (Partial Dependency)
FD5: s_sid → s_desc (Partial Dependency)

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)
PK: m_memberApplicationID
FK: m_gender references tb_ugender (ug_gid)
FK: m_religion references tb_ureligion (ua_rid)
FK: m_race references tb_urace (ur_rid)
FK: m_maritalStatus references tb_umaritalstatus (um_mid)

FK: m_status references tb_status (s_sid)
FK: m_homeState references tb_homestate (st_id)
FK: m_officeState references tb_officestatus (st_id)
FK: m_adminID references tb_admin (a_adminID)

tb_ugender (ug_gid, ug_desc)
PK: ug_gid

tb_umaritalstatus (um_mid, um_desc)
PK: um_mid

tb_urace (ur_rid, ur_desc)
PK: ur_rid

tb_ureligion (ua_rid, ua_desc)
PK: ua_rid

tb_homestate (st_id, st_desc)
PK: st_id

tb_offisestate (st_id, st_desc)
PK: st_id

tb_status (s_sid, s_desc)
PK: s_sid

tb_heir (h_heirID, h_memberApplicationID, h_name, hr_rid, hr_desc, h_ic)
PK: h_heirID
FK: h_memberApplicationID references tb_member (m_memberApplicationID)

Transitive Dependency:
FD2: hr_rid → hr_desc

tb_heir (h_heirID, h_memberApplicationID, h_name, h_relationWithMember, h_ic)
PK: h_heirID
FK: h_memberApplicationID references tb_member (m_memberApplicationID)
FK: h_relationWithMember references tb_hrelation (hr_rid)

tb_hrelation (hr_rid, hr_desc)
PK: hr_rid

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)
FK: f_memberNo references tb_member (m_memberNo)

No Transitive Dependency.

tb_banner (b_bannerID, b_name, b_banner, b_status, b_dateUpdated, b_adminID)

PK: b_bannerID

FK: b_adminID references tb_admin (a_adminID)

No Transitive Dependency.

tb_loan (l_loanApplicationID, l_memberNo, lt_lid, lt_desc, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, lb_id, lb_desc, l_monthlyNetSalary, l_signature, l_file, s_sid, s_desc, l_applicationDate, l_approvalDate, l_adminID)

PK: l_loanApplicationID

FK: l_memberNo references tb_member (m_memberNo)

FK: l_adminID references tb_admin (a_adminID)

Transitive Dependency:

FD1: lt_lid → lt_desc

FD2: lb_id → lb_desc

FD3: s_sid → s_desc

tb_loan (l_loanApplicationID, l_memberNo, l_loanType, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, l_bankName, l_monthlyNetSalary, l_signature, l_file, l_status, l_applicationDate, l_approvalDate, l_adminID)

PK: l_loanApplicationID

FK: l_loanType references tb_ltype (lt_lid)

FK: l_status references tb_status (s_sid)

FK: l_memberNo references tb_member (m_memberNo)

FK: l_bankName references tb_lbank (lb_id)

FK: l_adminID references tb_admin (a_adminID)

tb_ltype (lt_lid, lt_desc)

PK: lt_lid

tb_lbank (lb_id, lb_desc)

PK: lb_id

tb_status (s_sid, s_desc)

PK: s_sid

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)

PK: g_guarantorID

FK: g_loanApplicationID references tb_loan (l_loanApplicationID)

FK: g_memberNo references tb_member (m_memberNo)

No Transitive dependency.

tb_policies (p_policyID, p_memberRegFee, p_minShareCapital, p_minFeeCapital, p_minFixedSaving, p_minMemberFund, p_minMemberSaving, p_minOtherFees,

p_minShareCapitalForLoan, p_maxInstallmentPeriod, p_maxAlBai, p_rateAlInnah,
 p_maxAlInnah, p_rateAlInnah, p_maxBPulihKendaraan, p_rateBPulihKendaraan,
 p_maxCukaiJalanInsurans, p_rateCukaiJalanInsurans, p_maxKhas, p_rateKhas,
 p_maxKarnivalMusim, p_rateKarnivalMusim, p_maxAlQadrulHassan,
 p_rateAlQadrulHassan, p_salaryDeductionForSaving, p_salaryDeductionForMemberFund,
 p_dateUpdated, p_adminID)
 PK: p_policyID
 FK: p_adminID references tb_admin (a_adminID)

No Transitive dependency.

tb_transaction (t_transactionID, tt_id, tt_desc, t_transactionAmt, rm_id, rm_desc, t_year,
 t_desc, t_transactionDate, t_memberNo, t_adminID)
 PK: t_transactionID
 FK: t_memberNo references tb_member (m_memberNo)
 FK: t_adminID references tb_admin (a_adminID)

Transitive Dependency:

FD1: tt_id → tt_desc

FD2: rm_id → rm_desc

tb_transaction (t_transactionID, t_transactionType, t_transactionAmt, t_month, t_year,
 t_desc, t_transactionDate, t_memberNo, t_adminID)
 PK: t_transactionID
 FK: t_transactionType references tb_ttype (tt_id)
 FK: t_month references tb_rmonth (rm_id)
 FK: t_memberNo references tb_member (m_memberNo)
 FK: t_adminID references tb_admin (a_adminID)

tb_ttype (tt_id, tt_desc)
 PK: tt_id

tb_month (rm_id, rm_desc)
 PK: rm_id

3rd Normal Form (Rearranged)

tb_user (u_id, reset_token, token_expiry, u_pwd, u_type)

PK: u_id

FK: u_type references tb_utype (ut_tid)

tb_utype (ut_tid, ut_desc)

PK: ut_tid

tb_admin (a_adminID, a_name, a_ic, a_email, a_position, a_status, a_registrationDate)

FK: a_adminID references tb_user (u_id)

tb_member (m_memberApplicationID, m_name, m_ic, m_email, m_gender, m_religion, m_race, m_maritalStatus, m_homeAddress, m_homePostcode, m_homeCity, m_homeState, m_memberNo, m_pfNo, m_position, m_positionGrade, m_officeAddress, m_officePostcode, m_officeCity, m_officeState, m_phoneNumber, m_homeNumber, m_monthlySalary, m_feeMasuk, m_modalSyer, m_modalYuran, m_deposit, m_alAbrar, m_simpananTetap, m_feeLain, m_status, m_applicationDate, m_approvalDate, m_adminID)

PK: m_memberApplicationID

FK: m_gender references tb_ugender (ug_gid)

FK: m_religion references tb_ureligion (ua_rid)

FK: m_race references tb_urace (ur_rid)

FK: m_maritalStatus references tb_umaritalstatus (um_mid)

FK: m_status references tb_status (s_sid)

FK: m_homeState references tb_homestate (st_id)

FK: m_officeState references tb_officestate (st_id)

FK: m_adminID references tb_admin (a_adminID)

tb_ugender (ug_gid, ug_desc)

PK: ug_gid

tb_umaritalstatus (um_mid, um_desc)

PK: um_mid

tb_urace (ur_rid, ur_desc)

PK: ur_rid

tb_ureligion (ua_rid, ua_desc)

PK: ua_rid

tb_homestate (st_id, st_desc)

PK: st_id

tb_officestate (st_id, st_desc)

PK: st_id

tb_status (s_sid, s_desc)

PK: s_sid

tb_heir (h_heirID, h_memberApplicationID, h_name, h_relationWithMember, h_ic)

PK: h_heirID

FK: h_memberApplicationID references tb_member (m_memberApplicationID)

FK: h_relationWithMember references tb_hrelation (hr_rid)

tb_hrelation (hr_rid, hr_desc)

PK: hr_rid

tb_financial (f_memberNo, f_shareCapital, f_feeCapital, f_fixedSaving, f_memberFund, f_memberSaving, f_dateUpdated)

FK: f_memberNo references tb_member (m_memberNo)

tb_banner (b_bannerID, b_name, b_banner, b_status, b_dateUpdated, b_adminID)

PK: b_bannerID

FK: b_adminID references tb_admin (a_adminID)

tb_loan (l_loanApplicationID, l_memberNo, l_loanType, l_appliedLoan, l_loanPeriod, l_monthlyInstalment, l_loanPayable, l_bankAccountNo, l_bankName, l_monthlyNetSalary, l_signature, l_file, l_status, l_applicationDate, l_approvalDate, l_adminID)

PK: l_loanApplicationID

FK: l_loanType references tb_ltype (lt_lid)

FK: l_status references tb_status (s_sid)

FK: l_memberNo references tb_member (m_memberNo)

FK: l_bankName references tb_lbank (lb_id)

FK: l_adminID references tb_admin (a_adminID)

tb_ltype (lt_lid, lt_desc)

PK: lt_lid

tb_lbank (lb_id, lb_desc)

PK: lb_id

tb_guarantor (g_guarantorID, g_loanApplicationID, g_memberNo, g_signature)

PK: g_guarantorID

FK: g_loanApplicationID references tb_loan (l_loanApplicationID)

FK: g_memberNo references tb_member (m_memberNo)

tb_policies (p_policyID, p_memberRegFee, p_minShareCapital, p_minFeeCapital, p_minFixedSaving, p_minMemberFund, p_minMemberSaving, p_minOtherFees, p_minShareCapitalForLoan, p_maxInstallmentPeriod, p_maxAlBai, p_rateAlInnah, p_maxAlInnah, p_rateAlInnah, p_maxBPulihKenderaan, p_rateBPulihKenderaan, p_maxCukaiJalanInsurans, p_rateCukaiJalanInsurans, p_maxKhas, p_rateKhas, p_maxKarnivalMusim, p_rateKarnivalMusim, p_maxAlQadrulHassan, p_rateAlQadrulHassan, p_salaryDeductionForSaving, p_salaryDeductionForMemberFund, p_dateUpdated, p_adminID)

PK: p_policyID

FK: p_adminID references tb_admin (a_adminID)

tb_transaction (t_transactionID, t_transactionType, t_transactionAmt, t_month, t_year, t_desc, t_transactionDate, t_memberNo, t_adminID)

PK: t_transactionID

FK: t_transactionType references tb_ttype (tt_id)
FK: t_month references tb_rmonth (rm_id)
FK: t_memberNo references tb_member (m_memberNo)
FK: t_adminID references tb_admin (a_adminID)

tb_ttype (tt_id, tt_desc)

PK: tt_id

tb_month (rm_id, rm_desc)

PK: rm_id